

Behavioral analysis of electricity consumption characteristics for customer groups using the k-means algorithm

Ruobing Wu^{*}

Yunnan Power Grid Company Information Center, Kunming 650217, Yunnan, China

ARTICLE INFO

Keywords:

K-means algorithm
Electricity consumption characteristic
Behavior analysis
Yunnan power grid
Electricity customer

ABSTRACT

In the fierce competition of the electricity market, how to consolidate and develop customers is particularly important. Aiming to analyze the electricity consumption characteristics of customer groups, this paper used a k-means algorithm and optimized it. The number of clusters was determined by the Davies-Bouldin index (DBI). An improved Harris Hawks optimization (IHHO) algorithm was designed to realize the initial cluster center selection. Based on data such as electricity purchase and average electricity price, electricity customer groups were clustered using the IHHO-k-means algorithm. The IHHO-k-means algorithm achieved the best clustering effect on Iris, Wine, and Glass datasets compared with the traditional k-means and PSO-k-means algorithms. Taking Iris as an example, the optimal value of the IHHO-k-means algorithm was 96.538, with an accuracy rate of 0.932, precision and recall rates of 0.941 and 0.793, respectively, an F-measure of 0.861, and an area under the curve (AUC) value of 0.851. In the customer dataset, the number of clusters determined by DBI was 4. The power customers were divided into four groups with different characteristics of electricity consumption, and their electricity consumption behaviors were analyzed. The results prove the reliability of the IHHO-k-means algorithm in analyzing electricity consumption characteristics of customer groups, and it can be applied in practice.

1. Introduction

With the promotion of economic development, the competition in the electricity market is becoming more and more fierce [1], and the choice of customers is becoming more and more diversified. To avoid customer loss, electricity companies need to adjust their service strategies in time to better meet customer needs while ensuring the share of electricity sales and improving the competitiveness of enterprises. The promotion of the smart grid significantly enhances the ease of gathering and examining electricity consumption data of customer groups [2]. Analyzing customers can provide reliable support for adjusting and optimizing marketing strategies and service methods for electricity companies. Therefore, analyzing and utilizing electricity consumption data from the customer group is of great practical value to electricity companies.

With the advancement of data mining techniques, it has also been well applied in the electricity industry [3]. Currently, data mining in the power industry involves aspects such as electricity price prediction and load analysis. Johannesen et al. [4] conducted a study on load forecasting in rural microgrids in Norway through regression analysis and

found that regression analysis had good forecasting accuracy based on the data from 2014 to 2018. Alquthami et al. [5] analyzed the real-time electricity pricing policy and designed a model based on backtracking techniques to calculate electricity prices for various customer categories to estimate incentives through load-shifting profiles. They found that customers could gain some incentives through their involvement in the load scheduling process. Chen et al. [6] designed a hybrid method combining principal component analysis, Relief, and kernel extreme learning machine to forecast the level of icing on transmission lines and found that the method showed low error. Du et al. [7] clustered the load data of electricity users using the OCHNN-K-means algorithm and then proposed a time-sharing peak-valley electricity price method. A case study showed that it could effectively meet the different needs of customers. Alves et al. [8] analyzed the influence of energy produced under special regulations between 2009 and 2016 on the wholesale hourly spot market prices of electricity in Portugal, revealing a negative effect of the special regime on wholesale prices. Mohammadi et al. [9] studied the simulation of intraday electricity market prices using deep convolutional generative adversarial networks, long short-term memory, and the No-U-Turn sampler algorithm. They verified the applicability of this

^{*} Corresponding author.

E-mail address: wrb_rb@outlook.com.

<https://doi.org/10.1016/j.sasc.2024.200143>

Received 10 April 2024; Received in revised form 9 August 2024; Accepted 26 August 2024

Available online 29 August 2024

2772-9419/© 2024 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

method through price simulation in the Nordic intraday electricity market. Uniejewski [10] compared two new approaches to forecasting probabilistic electricity prices: the smooth quantile regression averaging (SQRA) and lasso quantile regression averaging. Through evaluation of a dataset from the Polish electricity market, it was found that both methods significantly improved the accuracy of predictions, with SQRA having a slight advantage. Beltrán et al. [11] proposed a statistical and machine learning-based model that incorporates time series analysis to achieve accurate price forecasting in the wholesale market, applied it to the Spanish wholesale market, and obtained good accuracy. Table 1 shows a summary of relevant literature.

Based on the above table, extensive research has been conducted on electricity price and load forecasting in the field of electricity market research. However, there is still limited research on customer segmentation analysis in the electricity market. Du et al. [7] clustered electricity load data using the k-means algorithm, providing support for peak-valley electricity pricing formulation. It can be found that clustering based on the k-means algorithm can also play a significant role in electricity market research, but it has not yet been fully utilized.

In view of the shortcomings of the k-means algorithm in electricity market research, this article analyzed the electricity consumption behavior of power customers using the improved k-means algorithm, further verifying its reliability in data clustering and providing

theoretical support for the application of more artificial intelligence methods in electricity market research. The k-means algorithm is a clustering analysis method that is simple and computationally efficient, and it is widely used in data analysis with good performance [12]. One major limitation of this approach lies in the challenge of estimating the cluster count and its susceptibility to variations in initial cluster centers. Therefore, this article proposes improvements on these two points. The research hypothesis of this paper is whether the improved k-means algorithm can classify different groups based on their electricity consumption characteristics among electricity customers.

The article provides a simple analysis of the electricity customer group's electricity usage characteristics in the second section. It introduces the k-means algorithm and explains the optimized k-means algorithm designed in this study. The experimental settings and evaluation criteria are presented. Chapter 3 presents the experimental results. In the fourth section, a discussion of these results is conducted, along with some suggestions for managing and marketing different customer groups. Conclusions are given in the fourth section.

2. Materials and methods

2.1. Behavioral analysis of electricity consumption characteristics of customer groups

In the face of existing customers and technologies, how to consolidate and expand market share and improve market influence has become a focus of major electricity companies. At present, electricity companies have different prices, products, and promotional activities for different customers. However, most of the current analysis of electricity customer groups is simply classified according to the user industry or the nature of electricity consumption, and the behavioral analysis of electricity characteristics is still less. With the development of smart grid, electricity companies can obtain the relevant data of electricity customer groups more conveniently and quickly, which provides favorable conditions for the behavioral analysis of different customer groups. Based on the analysis of customers' electricity characteristics, dividing them into different groups can help electricity companies provide more targeted power services. While consolidating existing customers, it can dig out customers with development value to realize the long-term development of electricity companies.

Yunnan Power Grid is responsible for the power supply of Yunnan province, covering 16 cities. This paper analyzed the electricity consumption characteristics of Yunnan power grid customer groups. Based on the current electricity market and actual sales situation, the customers' electricity consumption characteristics were considered from the following aspects.

- (1) Electricity consumption behavior
 - ① Electricity purchase: it indicates the electricity amount purchased by customers in the previous year.
 - ② Average electricity price: it reflects the carrying condition of customers' electrical equipment.
 - ③ Load rate: In general, new customers have a lower load rate.
 - ④ Coefficient of variation: it refers to the customer's electricity consumption volatility, the coefficient of variation = standard deviation of monthly electricity consumption/average monthly electricity consumption
- (2) Credit behavior
 - ① Electricity fee recovery rate: it refers to the recovery of electricity charges.
 - ② Amount of electricity stolen: it refers to the illegal amount of electricity consumed by customers.
- (3) Development behavior
 - ① Growth rate of electricity consumption: it denotes the increase in electricity usage by customers in the current year compared to the preceding year.

Table 1
Literature summary.

	Method	Result
Johannesen et al. [4]	Regression analysis	The joint learning of regression tools and autoregression could forecast the time-series components of various load distribution features.
Alquthami et al. [5]	Backtracking techniques	The algorithm converged to the optimal optimization result within a short period of time. The simulation found that users obtained some practical incentives by engaging in the load-scheduling process.
Chen et al. [6]	A hybrid method combining principal component analysis, Relief, and kernel extreme learning machine	It demonstrated high accuracy and low prediction error in real dataset predictions.
Du et al. [7]	OCHNN-K-means algorithm	Empirical analysis of actual data from a certain city revealed that formulating specific peak and off-peak electricity prices for different power users through this method reduced the electricity costs for demand-side power users and improved energy utilization efficiency.
Alves et al. [8]	Two-stage least square method and generalized method of moments multivariate regressions	The special regime negatively impacted the wholesale electricity spot prices.
Mohammadi et al. [9]	Deep convolutional generative adversarial networks, long short-term memory, and the No-U-Turn sampler algorithm	The method performed well in simulating prices in the Nordic intraday electricity market.
Uniejewski et al. [10]	SQRA and lasso quantile regression averaging	The two approaches significantly improved the accuracy of predictions on the Polish electricity market dataset, with SQRA showing a slight advantage.
Beltrán et al. [11]	An intelligent framework using a large number of statistical and machine learning models	The method demonstrated good average absolute error and average absolute scale error in the wholesale market in Spain.

② Electricity expansion quantity: when the customer expands the business, the capacity will be expanded, and the demand for expansion will be put forward to the electricity company.

2.2. The k-means algorithm and its improvement

2.2.1. The k-means algorithm

In the behavioral analysis of electricity consumption characteristics of customer groups, the k-means algorithm can categorize customers according to different electricity consumption characteristics. Electricity is a unique commodity, and electricity customers can be classified based on their electricity consumption scale, nature of usage, etc., to assist electricity companies in better managing their customers and implementing appropriate measures to retain, maintain, or attract them. The algorithm first clusters all the data into the nearest cluster, then updates the cluster center according to the mean value of the data in each cluster, and repeats until the criterion function converges. The criterion function is:

$$J = \sum_{i=1}^k \sum_{j=1}^n (c_i - x_j)^2 \quad (1)$$

where c_i stands for the cluster center of the i -th class and x_j stands for the j -th sample in the i -th class.

The k-means algorithm is a simple and fast method [13], which can classify the samples accurately. However, this method also has some defects:

- (1) the number of clusters needs to be determined according to prior knowledge [14];
- (2) the selection of the initial cluster center has a significant impact on its sensitivity [15].

2.2.2. The optimization of k-means

2.2.2.1. Davies-Bouldin index. In this paper, the number of clusters was determined using the Davies-Bouldin Index (DBI) [16], and the process is as follows.

- (1) The average distance between the j -th sample x_j in the i -th class L_i and cluster center c_i was defined as:

$$S_i : S_i = \frac{1}{|L_i|} \sum_{x_j \in L_i} \|x_j - c_i\|^2 \quad (2)$$

- (2) The distance between the i -th class and the j -th class was:

$$M_{ij} = \|c_i - c_j\|^2 \quad (3)$$

- (3) The DBI was calculated:

$$DBI(k) = \frac{1}{k} \sum_{i=1}^k \max_{j \neq i} \left(\frac{S_i + S_j}{M_{ij}} \right) \quad (4)$$

- (4) The DBI of each cluster number was calculated. The number of clusters was the best when the DBI value was the smallest.

2.2.2.2. Harris hawk optimization algorithm. The improved Harris hawk optimization algorithm (IHHO) was employed to determine the initial cluster centers. The HHO algorithm is derived from the hunting behavior of the Harris hawk [17], which has been successfully applied in various optimization issues [18]. The HHO algorithm takes Harris hawk as a candidate solution and prey as the optimal solution. Firstly, in the

exploration stage, the position update of the hawk is expressed as:

$$X(t+1) = \begin{cases} X_{rand}(t) - m_1 |X_{rand}(t) - 2m_2 X(t)|, p \geq 0.5 \\ [X_{rabbit}(t) - X_{ave}(t)] - m[LB + m_4(UB - LB)], p < 0.5 \end{cases} \quad (5)$$

$$X_{ave}(t) = \frac{1}{N} \sum_{i=1}^N X_i(t) \quad (6)$$

where $p \geq 0.5$ indicates that the hawk perches on a random high tree, $p < 0.5$ indicates that the hawk perches on a place near other individuals or preys, $X_{rand}(t)$ represents a random individual location, $X(t)$ stands for the individual position at the t -th iteration, $X_{rabbit}(t)$ stands for the prey position, $X_{ave}(t)$ is the average location of the current population, UB and LB are the upper and lower boundaries of the search domain, N is the population's size, m_1, m_2, m_3, m_4, p are random numbers obeying uniform distribution.

In the development phase, the hawk starts to hunt and the prey tries to escape. During the escape, the prey energy can be written as:

$$E = 2E_0(1 - t/T) \quad (7)$$

where E_0 denotes the initial value of prey energy and T stands for the maximum count of iterations.

A random number is used to describe that the prey can evade capture. $u < 0.5$ suggests the successful evasion of the prey, otherwise it means that the prey fails to escape. It is divided into the following situations.

- ① When $u \geq 0.5$ and $|E| \geq 0.5$, the position update of the hawk is:

$$X(t+1) = \Delta X(t) - E|JX_{rabbit}(t) - X(t)| \quad (8)$$

$$\Delta X(t) = X_{rabbit}(t) - X(t) \quad (9)$$

$$J = rand(0, 2) \quad (10)$$

where J is the random jump intensity when the prey escapes.

- ② When $u \geq 0.5$ and $|E| < 0.5$, the position update of the hawk is expressed as:

$$X(t+1) = X_{rabbit}(t) - E|\Delta X(t)| \quad (11)$$

- ③ When $u < 0.5$ and $|E| \geq 0.5$, the position update of the hawk is expressed as:

$$X(t+1) = \begin{cases} Y : X_{rabbit}(t) - E|JX_{rabbit}(t) - X(t)|, f(Y) < f[X(t)] \\ Z : Y + S \times Levy(D), f(Z) < f[X(t)] \end{cases} \quad (12)$$

where D is the dimension, S is a D -dimensional random vector, f is the fitness function, and $Levy$ is the Levy flight function.

- ④ When $u < 0.5$ and $|E| < 0.5$, the expression of the hawk's position update is:

$$X(t+1) = \begin{cases} Y : X_{rabbit}(t) - E|JX_{rabbit}(t) - X_{ave}(t)|, f(Y) < f[X(t)] \\ Z : Y + S \times Levy(D), f(Z) < f[X(t)] \end{cases} \quad (13)$$

2.2.2.3. Improved Harris hawk optimization algorithm. In the HHO algorithm, the initial quality of the population is poor, and the linear change of prey energy and jump intensity is easy to lead to local optimum [19]. Therefore, this paper improved this algorithm to obtain the IHHO algorithm. Firstly, a Tent map [20] is utilized to perturb the initial population, and the formula is:

$$X(t+1) = \begin{cases} X(t)/a, X(t) \in [0, a] \\ [1 - X(t)]/(1 - a), X(t) \in [a, 1] \end{cases} \quad (14)$$

where $a \in (0, 1)$.

Then, using nonlinear prey energy, E will be modified as:

$$E = 2E_0 \times \left(e^{-\frac{t+1}{T}} \right)^3 \quad (15)$$

Using nonlinear jump strength, J will be modified to:

$$J = 2r \left(e^{1 - \sqrt{\frac{t+1}{T}}} - 1 \right) \quad (16)$$

In the population initialization stage, the IHHO algorithm employs Tent mapping to enhance the variety of the population and broaden the search space. Subsequently, by making non-linear improvements in prey energy and jump strength, it enhances the algorithm's optimization capability and convergence performance. Consequently, it can achieve superior performance in optimizing initial cluster centers. The IHHO algorithm was adopted to optimize the initial clustering center of the k-means algorithm, and DBI was employed to identify the most suitable number of clusters to enhance the effectiveness of the k-means algorithm in electricity customer group clustering. The designed IHHO-k-means algorithm is displayed in Fig. 1.

2.3. Experimental setup

A 64-bit Windows 10 operating system with Python 3.9 as the programming language was used. The clustering effectiveness of the designed algorithm was verified on UCI datasets [21] (Table 2).

The data of the electricity customer group came from Yunnan Power Grid, and the data of 578 customers were collected. The data of each customer included eight characteristics, such as electricity purchase, average electricity price, etc., and they were normalized. Table 3 shows some of the data.

The clustering performance was evaluated, as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (17)$$

$$Precision = \frac{TP}{TP + FP} \quad (18)$$

$$Recall = \frac{TP}{TP + FN} \quad (19)$$

Table 2

UCI datasets.

Dataset	Number	Feature dimension	Number of clusters
Iris	150	4	3
Wine	178	13	3
Glass	214	9	6

$$F - measure = \frac{2Precision \cdot Recall}{Precision + Recall} \quad (20)$$

The value of the area under the curve (AUC): The area under the receiver operating characteristic (ROC) curve in a coordinate system with false positive rate (FPR) as the horizontal axis and true positive rate (TPR) as the vertical axis:

$$FPR = \frac{FP}{TN + FP} \quad (21)$$

$$TPR = \frac{TP}{TP + FN} \quad (22)$$

In the above equations, TP indicates the quantity of samples correctly classified as positive, TN indicates the quantity of samples correctly classified as negative, FP indicates the quantity of samples that are misclassified as positive, and FN is the quantity of samples that were misclassified as negative.

Using the UCI dataset, each algorithm being compared was run 30 times, with the best value being the optimal value among the 30 tests. The accuracy and F-measure were averaged.

3. Results and analysis

The following algorithms were compared using Iris, Wine, and Glass:

- (1) the traditional k-means algorithm,
- (2) K-medoids [22],
- (3) the particle swarm optimization (PSO) algorithm-based k-means algorithm [23],
- (4) the k-means algorithm using the grey wolf optimization (GWO) algorithm [24],
- (5) the HHO-k-means algorithm,

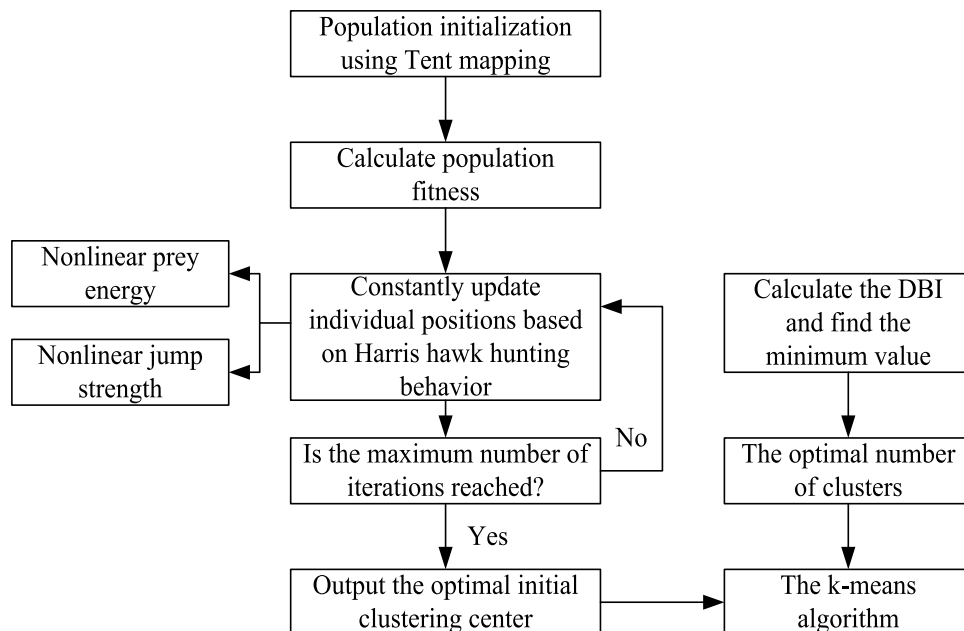


Fig. 1. The flowchart of the IHHO-k-means algorithm.

Table 3
Electricity customer group data.

Customer number	001	002	003	004	005	...
Electricity purchase/kWh	12,121,411	12,012,154	6582,145	2545,185	256,854	...
Average electricity price/(yuan/kWh)	0.68	0.75	0.64	0.81	0.77	...
Load rate	0.61	0.56	0.60	0.55	0.62	...
Coefficient of variation	0.212	0.384	0.485	0.155	0.164	...
Electricity fee recovery rate/%	97.26	85.33	87.68	77.55	91.26	...
Amount of electricity stolen/kw	0	0	0	0	0	...
Growth rate of electricity consumption/%	3.32	25.61	200.33	1.26	3.25	...
Electricity expansion quantity	0	0	1	0	0	...

(6) the IHHO-k-means algorithm.

Table 4 presents the comparison results.

It was observed that the conventional k-means algorithm had the worst clustering performance, with the largest best value and the lowest accuracy and F-measure, and the clustering performance of k-medoids was average. In comparison, the IHHO-k-means algorithm achieved the best clustering performance for the three datasets. For the Iris dataset, the best value of the IHHO-k-means algorithm was 96.538, the accuracy was 0.932, the precision and recall rate were 0.941 and 0.793, the F-measure was 0.861, and the AUC value was 0.851. For the Wine dataset, the best value of the IHHO-k-means algorithm was 16,292.186, the accuracy was 0.757, the precision and recall rate were 0.791 and 0.568, the F-measure was 0.595, and the AUC value was 0.696. For the Glass dataset, the best value of this algorithm was 215.059, the accuracy was 0.739, the precision and recall rate were 0.672 and 0.537, the F-measure was 0.597, and the AUC value was 0.631.

Taking the Iris dataset as an example, the cluster convergence curves of different methods are displayed in Fig. 2.

From Fig. 2, it can be observed that among the several methods compared, the IHHO-k-means algorithm exhibited the best convergence accuracy and efficiency. Compared to other algorithms, the IHHO-k-means algorithm converged faster and more steadily. These findings suggested that the IHHO algorithm had superior performance in selecting initial clustering centers for the k-means algorithm and could effectively improve its clustering effectiveness and convergence performance.

The IHHO-k-means algorithm was applied to the clustering of electricity customer groups. First, the number of clusters was determined according to the DBI (Fig. 3).

According to the DBI, the optimal clustering performance was observed with a cluster count of 4. According to the different characteristics of electricity consumption, the customers were divided into four groups. In different groups, some features exhibiting significant

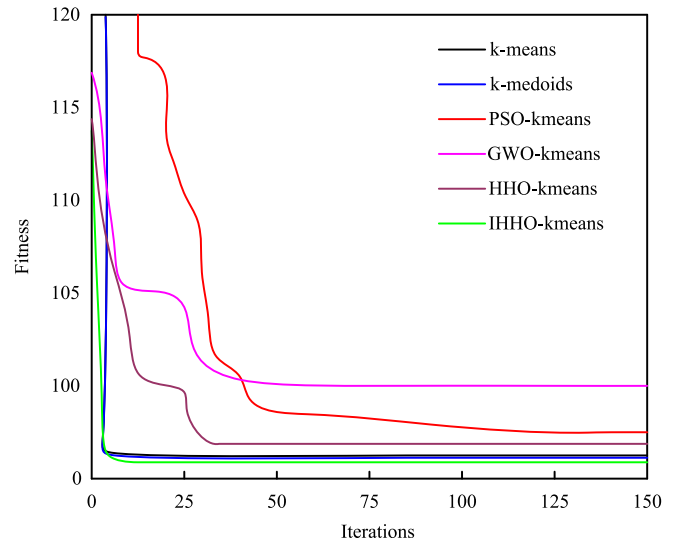


Fig. 2. The cluster convergence diagram for Iris.

distinctiveness are displayed in Table 5.

The results in Table 5 are presented in the form of Fig. 4.

According to Fig. 4, the analysis of the four groups in Table 4 is as follows.

(1) Group 1: High-value customers

The characteristics of electricity consumption behaviors of this group were analyzed, which were mainly manifested as large electricity purchase, low electricity price, high growth rate of electricity consumption, small volatility of electricity consumption (coefficient of variation), and positive electricity charge

Table 4
Comparison results of clustering performance.

		Best value	Accuracy	Precision	Recall rate	F-measure	AUC
Iris	k-means	98.356	0.874	0.886	0.759	0.818	0.827
	K-medoids	98.267	0.875	0.889	0.762	0.821	0.829
	PSO-k-means	98.141	0.877	0.892	0.768	0.825	0.831
	GWO-k-means	97.964	0.909	0.911	0.771	0.835	0.833
	HHO-k-means	96.871	0.921	0.931	0.782	0.850	0.845
	IHHO-k-means	96.538	0.932	0.941	0.793	0.861	0.851
Wine	k-means	16,555.684	0.711	0.764	0.551	0.640	0.681
	K-medoids	16,521.517	0.713	0.766	0.555	0.644	0.685
	PSO-k-means	16,451.258	0.715	0.771	0.558	0.647	0.688
	GWO-k-means	16,325.214	0.722	0.781	0.561	0.653	0.691
	HHO-k-means	16,303.225	0.737	0.783	0.564	0.656	0.693
	IHHO-k-means	16,292.186	0.757	0.791	0.568	0.661	0.696
Glass	k-means	222.676	0.636	0.657	0.512	0.576	0.612
	K-medoids	220.852	0.641	0.659	0.515	0.578	0.615
	PSO-k-means	218.651	0.656	0.661	0.522	0.583	0.621
	GWO-k-means	217.022	0.677	0.665	0.526	0.587	0.625
	HHO-k-means	216.841	0.708	0.669	0.529	0.591	0.628
	IHHO-k-means	215.059	0.739	0.672	0.537	0.597	0.631

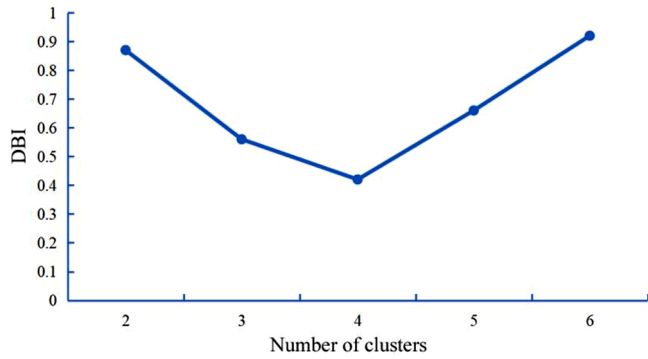


Fig. 3. DBI and the number of clusters.

Table 5
Clustering results of electricity customer groups3.

Electricity customer group	1	2	3	4
Proportion	4.21	10.64	68.99	16.16
Electricity purchase	2.3251	0.1426	1.1685	0.1256
Average electricity price	0.7123	0.1251	1.8562	0.0745
Load rate	0.5162	0.2125	0.1851	0.1511
Growth rate of electricity consumption	0.2325	0.8745	0.0125	2.3215
Electricity fee recovery rate	0.0158	0.1251	0.0215	0.1168
Coefficient of variation	0.0121	0.8326	0.0135	0.5652

recovery rate although not the highest, indicating that this group can create great value for the electricity company, so they are high-value customers.

(2) Group 2: high-potential customers

The characteristics of the electricity consumption behavior of this group were analyzed, and the results were small electricity

purchases, high growth rate of electricity consumption, and large fluctuation of electricity consumption. Although the current purchase of electricity of this group is small, the growth rate of electricity consumption is very high, and the coefficient of variation is also very high, indicating that this group is in the process of rapid industry growth, and its demand for electricity is large. It can be believed that such groups will bring great value to electricity companies in the future, so they are high-potential customers.

(3) Group 3: ordinary customers

By analyzing the characteristics of electricity consumption behaviors of this group, it was found that the purchase of electricity was medium, the electricity price was high, the growth rate of electricity consumption was low, and the fluctuation of electricity consumption was small. It is the main part of the Yunnan power grid customer group. Although the potential for future development is small, due to the large base, they can be regarded as ordinary customers.

(4) Group 4: low-value customers

By analyzing the characteristics of electricity consumption behaviors of this group, it was found that the purchase of electricity, the growth rate of electricity consumption, and the recovery rate of electricity consumption were all low, and the fluctuation of electricity consumption was large, indicating that this group is very unstable. There are more cases of not paying on time, and there is the phenomenon of customer loss, indicating low development potential.

4. Discussion

This article adopted the k-means algorithm to study the electricity consumption behavior of customers. Improvements were made to the

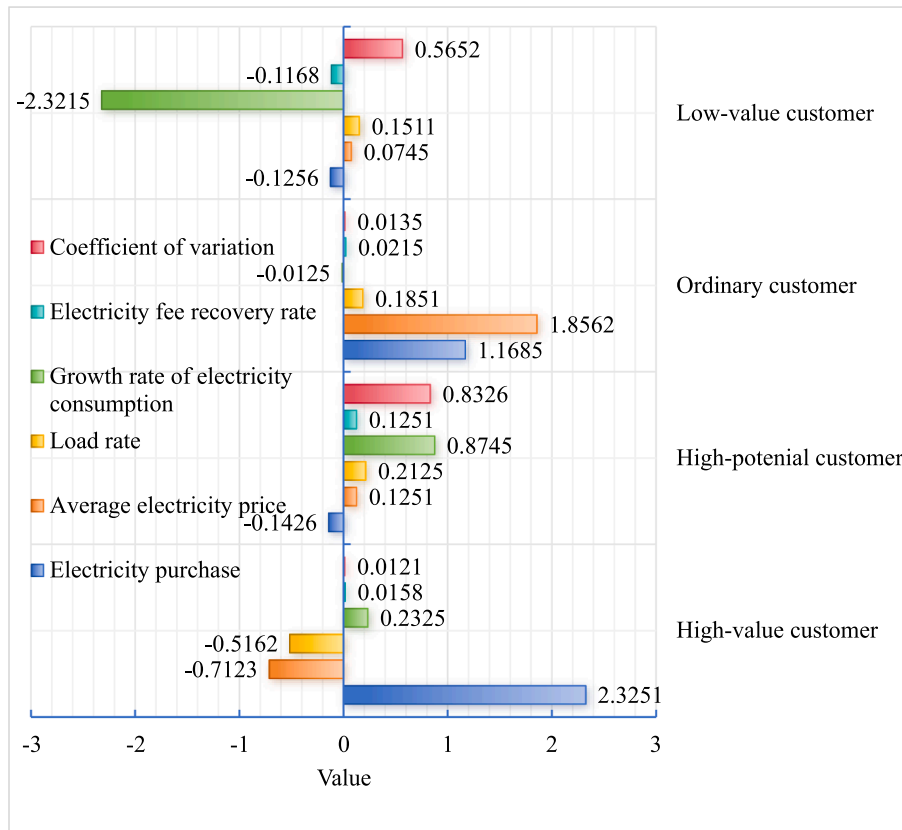


Fig. 4. Analysis of clustering results of electricity customer groups.

conventional k-means algorithm in terms of the number of clusters and initial cluster centers, resulting in the IHHO-k-means algorithm. It was found that the clustering effectiveness of the IHHO-k-means algorithm. Subsequently, a clustering analysis was conducted on a real electricity customer dataset from Yunnan Power Grid.

The results on the UCI dataset showed that the k-means algorithm optimized by an intelligent algorithm had better clustering performance compared to traditional k-means and k-medoids. It exhibited higher values in terms of indicators such as the best value and accuracy. Among various methods, the IHHO-k-means algorithm achieved the best results, proving the reliability of improvements to the k-means algorithm.

According to the clustering results, the electricity customers of Yunnan Power Grid were divided into four different groups based on their different electricity usage behaviors. For each group, the following management and marketing strategies can be adopted.

- (1) High-value customers: provide customers with exclusive quality services and give priority to value-added services; appoint account managers to timely grasp the changes in customers' electricity demand and adjust the marketing direction; provide certain preferential services according to the characteristics of customers' electricity consumption to improve the enthusiasm of customers' payment.
- (2) High-potential customers: provide high-quality services while thoroughly analyzing customer needs, excavate the potential demands for electricity, and meet them accordingly; maintain close contact with customers, improve customer care, and actively provide maintenance, consultation, and other services; study the law of customer electricity consumption and provide electricity consumption analysis reports to improve customer satisfaction and stickiness.
- (3) Ordinary customers: provide business guidance according to customer needs and conduct power supply reliability analysis to improve customer loyalty; provide customers with electricity consumption scale and efficiency analysis services to help customers improve equipment utilization; design a variety of plans through a pricing strategy to encourage customers to increase the purchase of electricity.
- (4) Low-value customers: try to help customers improve their electricity efficiency while providing basic services, and promote this group to change to type 1 or type 2; focus on customers who are in arrears many times and regularly return to visit and evaluate them to reduce the risk of electricity charge recovery; provide paid maintenance services to ensure the safe use of electricity.

5. Conclusions

In this paper, the k-means algorithm was utilized to analyze the electricity consumption characteristics of customer groups, and the conventional k-means algorithm was optimized to optimize its clustering performance. The determination of the number of clusters was based on DBI, and the initial cluster centers were determined by the IHHO algorithm.

Through the verification of different datasets, it was found that the IHHO-k-means algorithm had excellent clustering performance. In the electricity customer data, the customers were divided into four groups according to different characteristics of electricity consumption, and the consumption behavior and future service methods of different groups were analyzed, which provides some reliable support for the better development of the electricity company.

The research in this paper demonstrates the reliability of the IHHO-k-means algorithm in data clustering and proves the usefulness of the proposed method in behavior analysis of electricity consumption characteristics among customers. However, there are still some limitations, such as limited actual electricity consumption data collected from customer groups and a lack of validation for recommendations made for

different groups in practical applications.

In future work, further verification of the proposed method will be conducted on more comprehensive datasets, and an analysis will be performed to evaluate its effectiveness when applied in actual power companies, providing more robust support for its promotion and application.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CRediT authorship contribution statement

Ruobing Wu: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] J. Han, S. Lan, Q. Zhu, S. Yu, A study on the competition of electricity sellers in the open electricity sales market, in: 2020 5th Asia Conference on Power and Electrical Engineering (ACPEE), 2020, pp. 1028–1032, <https://doi.org/10.1109/ACPEE48638.2020.9136450>.
- [2] Z. Bayasgalan, T. Bayasgalan, Applications of smart customer model for smart grid, in: 2020 IEEE Region 10 Symposium (TENSYP), 2020, pp. 543–546, <https://doi.org/10.1109/TENSYP50017.2020.9231016>.
- [3] X. Huang, R. Fan, X. Guo, J. Chen, L. Guo, Z. Yuan, K. Wang, J. Zhang, Research on distributed photovoltaic power prediction based on centralized photovoltaic output, IOP Conf. Ser.: Earth Environ. Sci. 467 (2020) 1–8, <https://doi.org/10.1088/1755-1315/467/1/012080>.
- [4] N.J. Johannesen, M.L. Kolhe, M. Goodwin, Smart load prediction analysis for micro-grid with Holiday Cabins in Norwegian rural area 1, J. Clean. Prod. 266 (2020) 121423, <https://doi.org/10.1016/j.jclepro.2020.121423>.
- [5] T. Alquthami, A.H. Milyani, M. Awais, M.B. Rasheed, An incentive based dynamic pricing in smart grid: a customer's perspective, Sustainability 13 (2021) 1–17, <https://doi.org/10.3390/su13116066>.
- [6] Y. Chen, J. Fan, Z. Deng, B. Du, X. Huang, Q. Gui, PR-KELM: icing level prediction for transmission lines in smart grid, Future Gener. Comp. Sy. 102 (2020) 75–83, <https://doi.org/10.1016/j.future.2019.08.002>.
- [7] P. Du, D. Huang, Z. Liu, H. Zhang, J. Zhang, Q. Yan, Power customer classification based on OCHNN-Kmeans and price setting of TOU, in: 2020 5th Asia Conference on Power and Electrical Engineering (ACPEE), 2020, pp. 252–256, <https://doi.org/10.1109/ACPEE48638.2020.9136398>.
- [8] C.F. Alves, P.D. Pinto, Impact of special regime generation management on electricity prices: the Portuguese case, Int. J. Energy Sect. Ma. 16 (2022) 511–528, <https://doi.org/10.1108/IJESM-04-2020-0007>.
- [9] S. Mohammadi, M.R. Hesamzadeh, Econometric modeling of intraday electricity market price with inadequate historical data, in: 2022 IEEE Workshop on Complexity in Engineering (COMPENG), 2022, pp. 1–9, <https://doi.org/10.1109/COMPENG50184.2022.9905434>.
- [10] B. Uniejewski, Enhancing accuracy of probabilistic electricity price forecasting: a comparative study of novel quantile regression averaging generalization, in: 2023 19th International Conference on the European Energy Market (EEM), 2023, pp. 1–5, <https://doi.org/10.1109/EEM58374.2023.10161748>.
- [11] S. Beltrán, A. Castro, I. Irizar, G. Naveran, I. Yeregui, Framework for collaborative intelligence in forecasting day-ahead electricity price, Appl. Energ. 306 (2022) 1–25, <https://doi.org/10.1016/j.apenergy.2021.118049>.
- [12] J. Natarajan, B. Rebekka, An energy efficient dynamic small cell on/off switching with enhanced k-means clustering algorithm for 5G HetNets, Int. J. Commun. Netw. Di. 29 (2023) 209–237, <https://doi.org/10.1504/ijcnds.2023.10047234>.
- [13] B. Biantoro, Hernadewita, K-means clustering on quality of radial run out tires, IOP Conf. Ser.: Mater. Sci. Eng. 1034 (2021) 012122, <https://doi.org/10.1088/1757-899X/1034/1/012122>.

- [14] A. Pugazhenthil, L.S. Kumar, Selection of optimal number of clusters and centroids for k-means and fuzzy c-means clustering: a review, in: 2020 5th International Conference on Computing, Communication and Security (ICCCS), 2020, pp. 1–4, <https://doi.org/10.1109/ICCCS49678.2020.9276978>.
- [15] Y. Xiong, Q. Peng, Z. Zhang, Research on MapReduce parallel optimization method based on improved k-means clustering algorithm, in: DSIT 2020: 2020 3rd International Conference on Data Science and Information Technology, 2020, pp. 47–52, <https://doi.org/10.1145/3414274.3414282>.
- [16] A.K. Singh, S. Mittal, P. Malhotra, Y.V. Srivastava, Clustering evaluation by davies-bouldin index (DBI) in cereal data using k-means, in: 2020 Fourth International Conference on Computing Methodologies and Communication (ICCMC), 2020, pp. 306–310, <https://doi.org/10.1109/ICCMC48092.2020.ICCMC-00057>.
- [17] E.M. Golafshani, M. Arashpour, A. Behnood, Predicting the compressive strength of green concretes using Harris hawks optimization-based data-driven methods, Constr. Build. Mater. 318 (2022) 1–13, <https://doi.org/10.1016/j.conbuildmat.2021.125944>.
- [18] T. Thaher, M. Saheb, H. Turabieh, H. Chantar, Intelligent detection of false information in arabic tweets utilizing hybrid harris hawks based feature selection and machine learning models, Symmetry (Basel) 13 (2021) 1–24, <https://doi.org/10.3390/sym13040556>.
- [19] W. Long, J. Jiao, M. Xu, M. Tang, T. Wu, S. Cai, Lens-imaging learning Harris hawks optimizer for global optimization and its application to feature selection, Expert Syst. Appl. 202 (2022) 1–21, <https://doi.org/10.1016/j.eswa.2022.117255>.
- [20] S. Chen, S. Wang, An optimization method for an integrated energy system scheduling process based on NSGA-II improved by tent mapping chaotic algorithms, Processes 8 (2020) 1–11, <https://doi.org/10.3390/pr8040426>.
- [21] UC Irvine Machine Learning Repository, 2024. <https://archive.ics.uci.edu/datasets>.
- [22] N.I. Asriny, D.R. Ramadhanty, M.D. Izzati, M.L.P. Putra, A. Shofura, C.U. Albi, E. Widodo, Comparison of k-medoids and self organizing maps algorithm in grouping hydrometeorological natural disasters in Java Island, IOP Conf. Ser. Mater. Sci. Eng. 1077 (2021) 1–13, <https://doi.org/10.1088/1757-899X/1077/1/012008>.
- [23] H. Zhang, Q. Peng, PSO and K-means-based semantic segmentation toward agricultural products, Future Gener. Comput. Syst. 126 (2022) 82–87, <https://doi.org/10.1016/j.future.2021.06.059>.
- [24] N. Bharanidharan, R. Harikumar, Modified grey wolf randomized optimization in dementia classification using MRI images, IETE J. Res. 68 (2020) 1–10, <https://doi.org/10.1080/03772063.2020.1715852>.